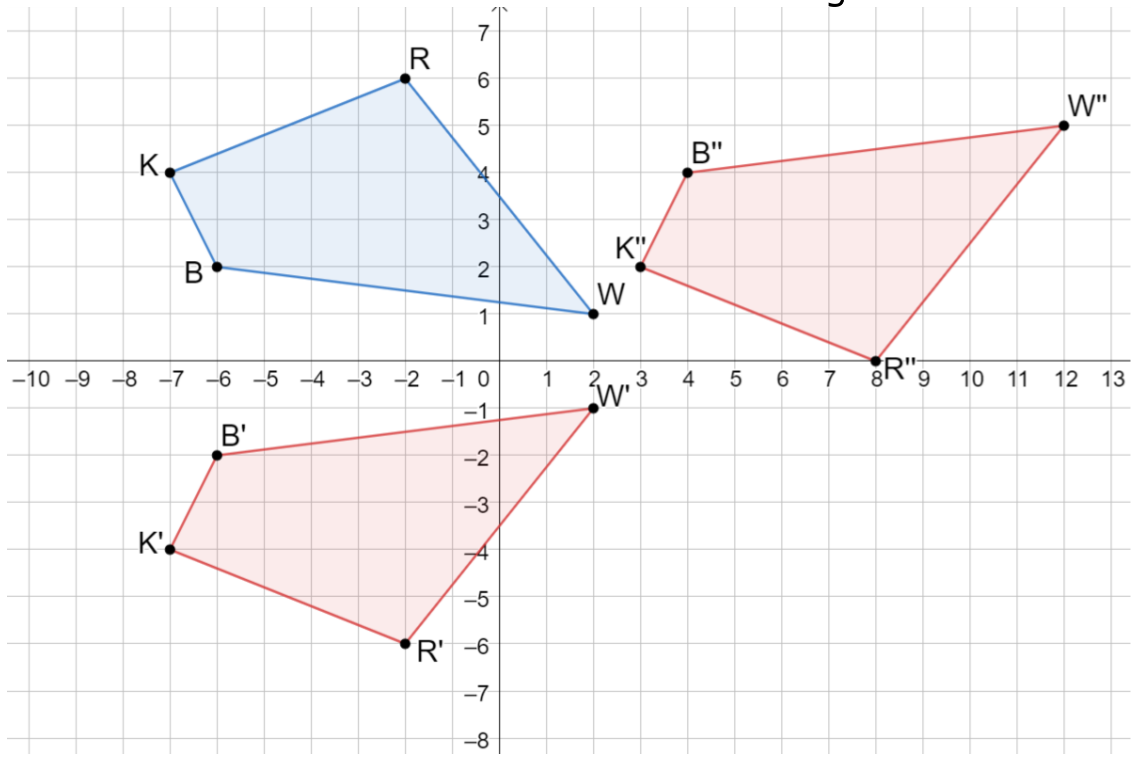
Geometry
Unit 4 Assessment

Name____________________
Date__________________
Period__________

1. Quadrilateral $KRWB$ is shown on the coordinate grid.



a. Apply the sequence of transformations to quadrilateral $KRWB$. Draw the result of each transformation on the coordinate grid.

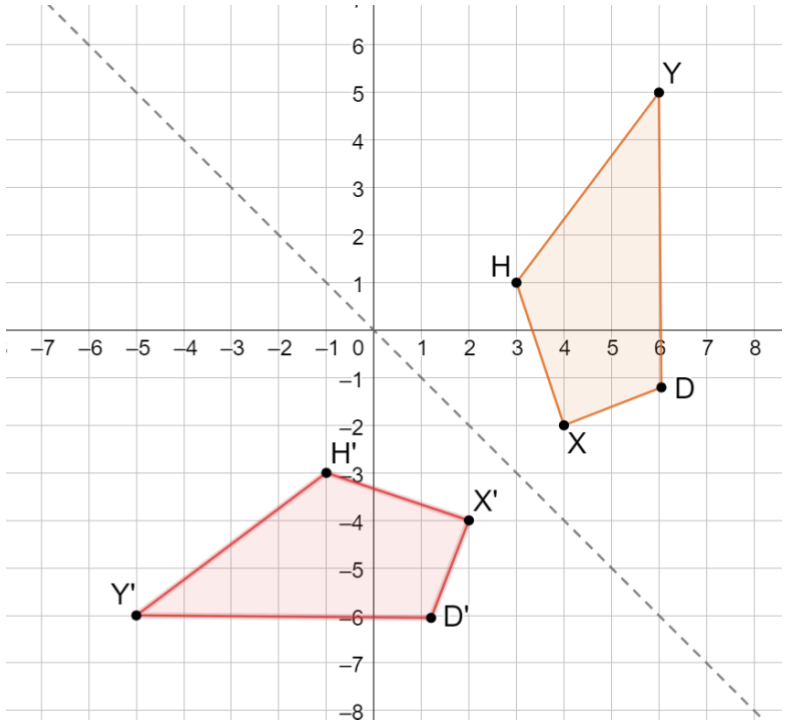
- $(x, y) \rightarrow (x, -y)$ Label as $K'R'W'B'$.
- $(x, y) \rightarrow (x + 10, y + 6)$ Label as $K''R''W''B''$.

b. Explain how the sequence of transformations for quadrilateral $KRWB$ resulted in $KRWB \cong K''R''W''B''$.

Answers will vary. Possible responses are shown.

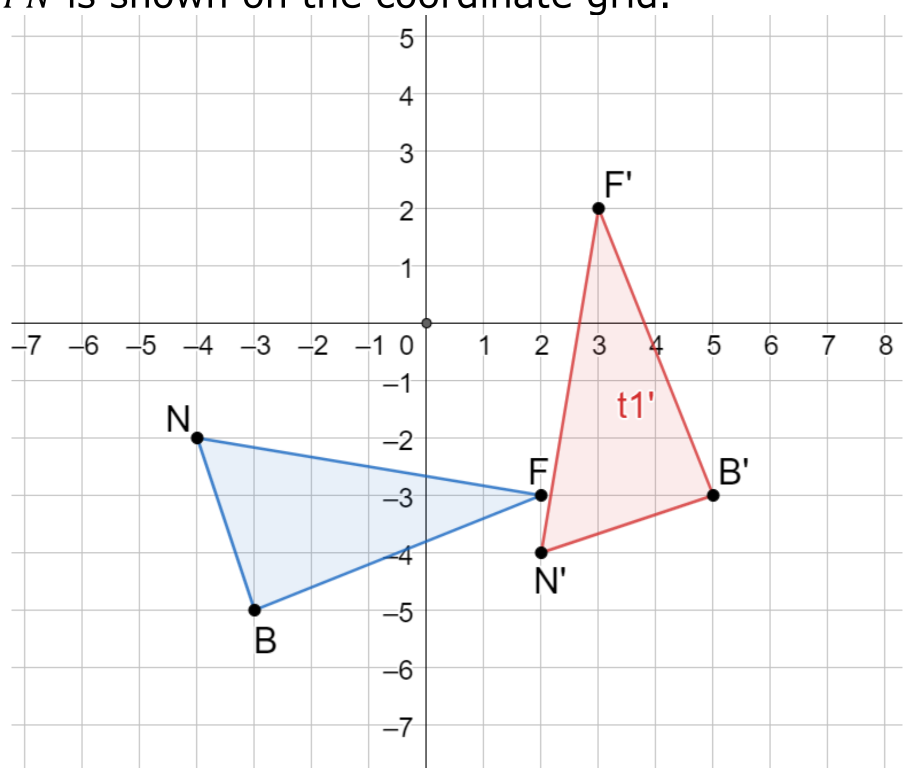
- *The first transformation is a reflection, and the second transformation is a translation. Both transformations are rigid motions which preserve both distance and angle measure.*
- *All points on quadrilateral* $KRWB$ *can be mapped to all points on* quadrilateral $K'R'W'B'$ because $(x, y) \rightarrow (x, -y)$ is a rigid motion. *All points on quadrilateral* $K'R'W'B'$ *can be mapped to all points on* quadrilateral $K''R''W''B''$ because $(x, y) \rightarrow (x + 10, y + 6)$ is a rigid motion.

2. Quadrilateral $YDXH$ and the line $y = -x$ are shown on the coordinate grid.



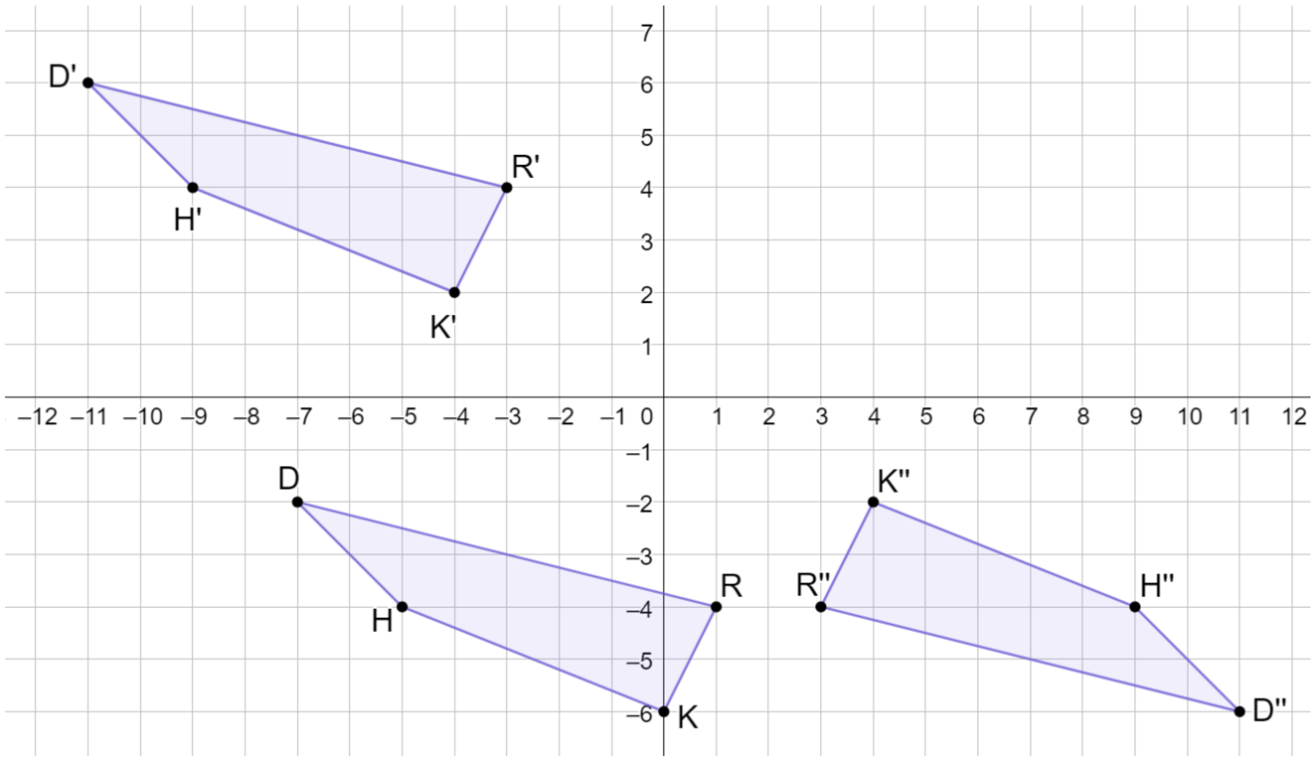
Reflect quadrilateral $YDXH$ across the line $y = -x$. Draw the result of the reflection on the coordinate grid.

3. Triangle BFN is shown on the coordinate grid.



Rotate ΔBFN $90°$ counterclockwise about the origin. Draw the result of the rotation on the coordinate grid.

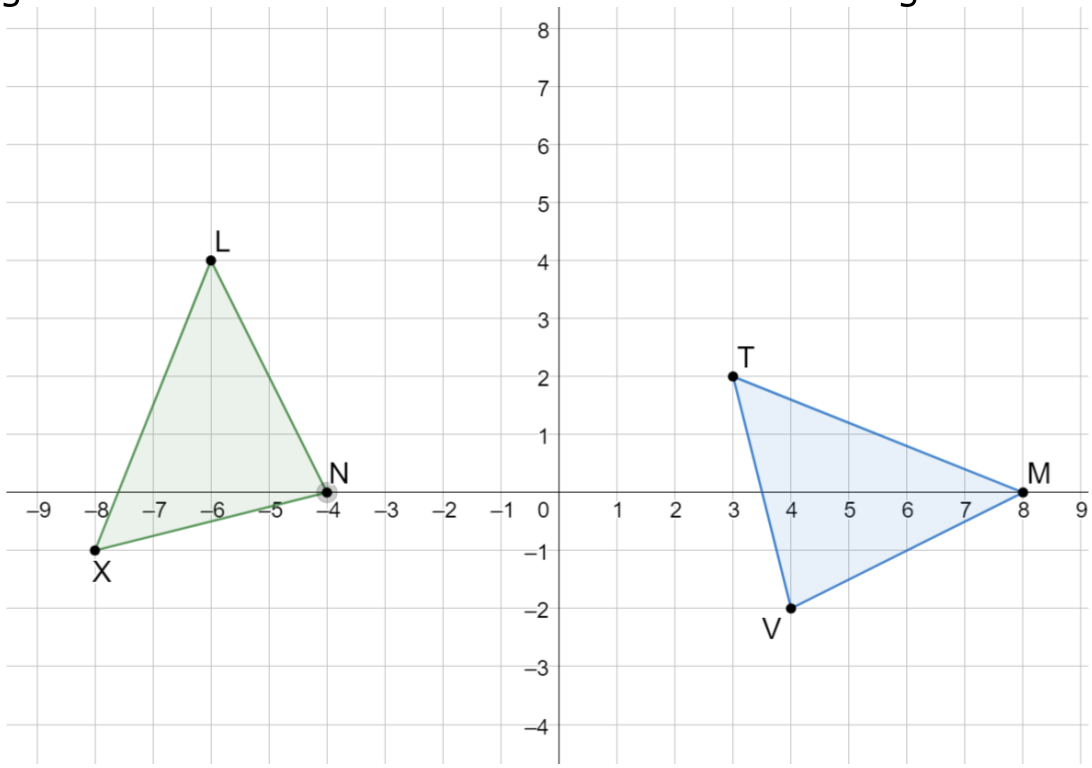
4. Quadrilateral $RKHD$ was transformed to create quadrilateral $R'K'H'D'$. Then, quadrilateral $R'K'H'D'$ was transformed to create quadrilateral $R''K''H''D''$. The quadrilaterals are shown on the coordinate grid.



Complete the table by writing a description for each of the transformations.

	Quadrilateral $RKHD$ Transformed to Quadrilateral $R'K'H'D'$	Quadrilateral $R'K'H'D'$ Transformed to Quadrilateral $R''K''H''D''$
Written Description	*Quadrilateral $RKHD$ is translated left 4 units and up 8 units to create* quadrilateral $R'K'H'D'$.	Quadrilateral $R'K'H'D'$ is rotated $180°$ clockwise about the origin to create quadrilateral $R''K''H''D''$*.*

5. Triangles TMV and LNX are shown on the coordinate grid.



a. Complete the statement.

The sequence of transformations that will map ΔTMV onto ΔXLN

is first

- ○ translate ΔTMV left 11, up 1,
- ○ reflect ΔTMV across the y-axis,
- ○ rotate ΔTMV $90°$ clockwise about the origin,
- ○ rotate ΔTMV $90°$ counterclockwise about the origin,

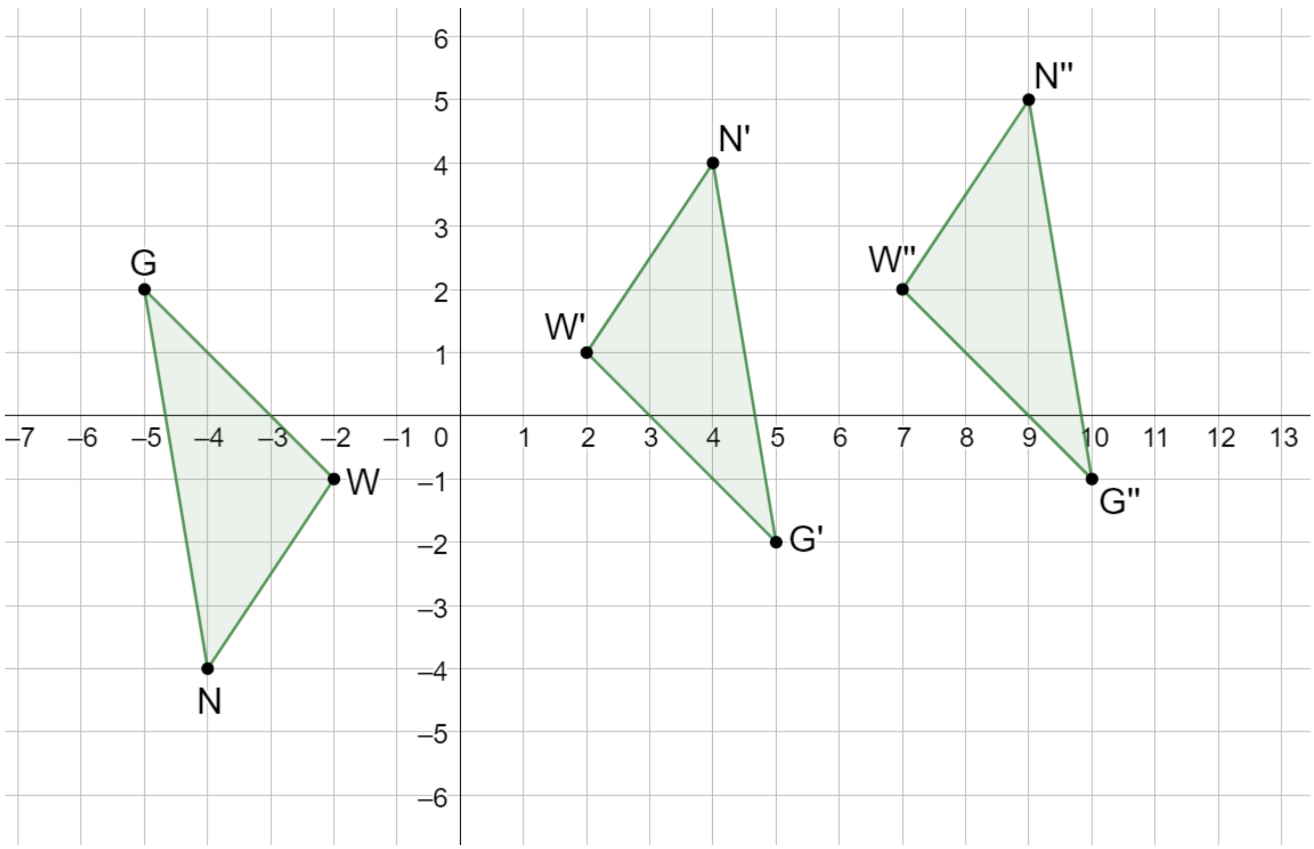
then

- ○ translate $\Delta T'M'V'$ left 3, up 2.
- ○ translate $\Delta T'M'V'$ left 2, down 3.
- ○ translate $\Delta T'M'V'$ left 6, down 4.
- ○ reflect $\Delta T'M'V'$ across the $y = -2$.
- ○ rotate ΔTMV $90°$ counterclockwise about the origin.

b. Complete the statements by filling in the blanks.

Vertex M maps to vertex _L___, vertex T maps to vertex ___ X _, and vertex V maps to vertex _ N ___. Therefore, ΔTMV is congruent to Δ_XLN__ by the definition of congruence in terms of rigid motions.

6. Triangles NWG, $N'W'G'$, and $N''W''G''$ are shown on the coordinate grid, where $\Delta N'W'G'$ and $\Delta N''W''G''$ are transformations of ΔNWG.



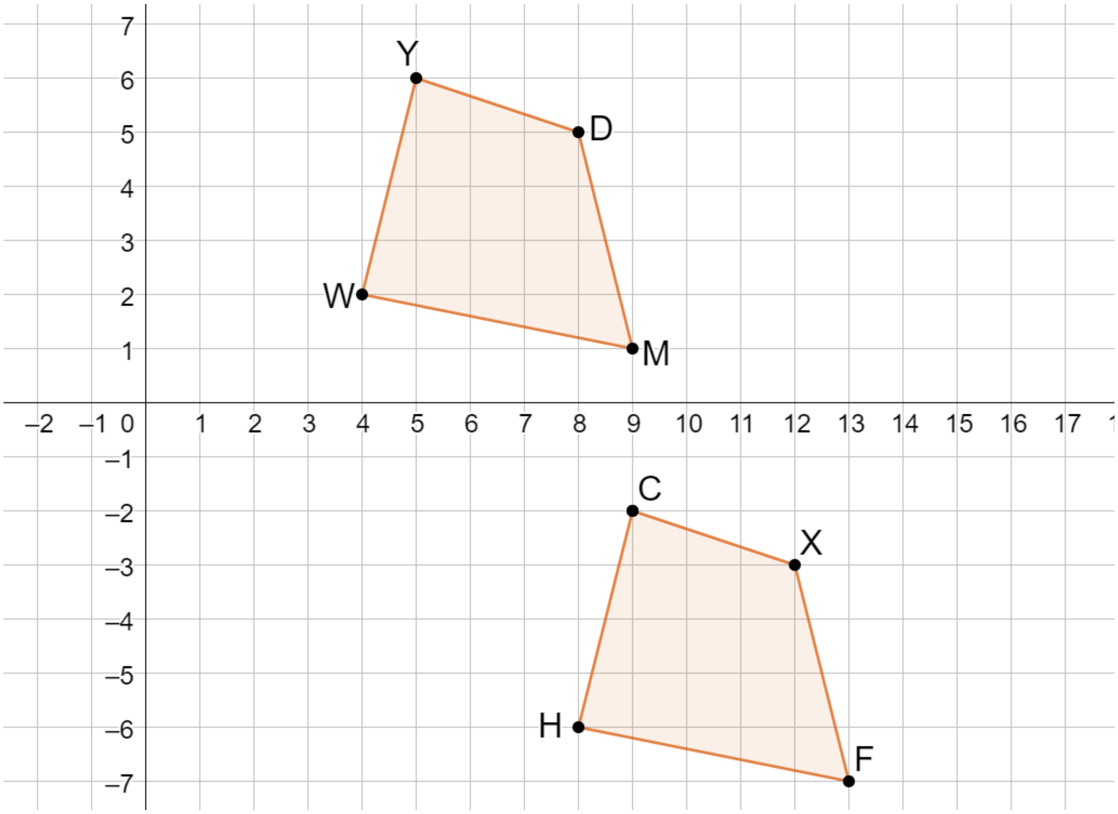
a. Describe the feature(s) of the transformations of ΔNWG that guarantees that each side length of ΔNWG is preserved for this sequence of transformations.

Answers will vary: Side length is preserved because the transformations are rigid motions and rigid motions preserve distance.

b. Determine a sequence of transformations so that when applied, ΔNWG will map onto $\Delta N''W''G''$. Complete the table by writing the algebraic description using coordinates for each mapping.

Mapping	**Algebraic Description**
$\Delta NWG \rightarrow \Delta N'W'G'$	$(x, y) \rightarrow (-x, -y)$
$\Delta N'W'G' \rightarrow \Delta N''W''G''$	$(x, y) \rightarrow (x + 5, y + 1)$

7. Quadrilateral $WYDM$ was transformed to create quadrilateral $HCXF$.



a. Write the algebraic description using coordinates to describe the transformation of quadrilateral $WYDM$ used to create quadrilateral $HCXF$.

Algebraic Description	$(x, y) \rightarrow (x + 4, y - 8)$

b. Use the algebraic description and the vertices of quadrilateral $WYDM$ as inputs to show the outputs that result in the vertices of quadrilateral $HCXF$.

Vertex of $WYDM$	**Work to determine the x-coordinate of $HCXF$**	**Work to determine the y-coordinate of $HCXF$**
W	$4 + 4 = 8$	$2 - 8 = -6$
Y	$5 + 4 = 9$	$6 - 8 = -2$
D	$8 + 4 = 12$	$5 - 8 = -3$
M	$9 + 4 = 13$	$1 - 8 = -7$